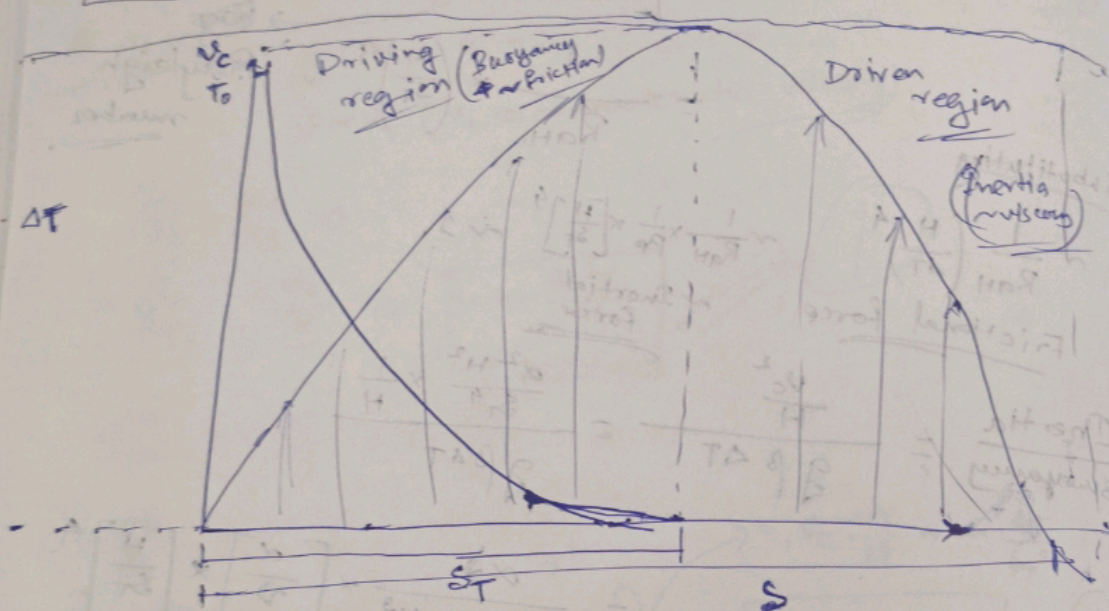


Energy Equation

$$v_c \sim \frac{\alpha H}{S_T^2}$$

$$v_c \sim \frac{\alpha}{H} Ra_H^{1/2}$$



Inertia $\rightarrow$ Driving the fluid $x \rightarrow$ (far away from wall)
 viscous $\rightarrow$ Resist the flow of fluid.

Driven region $\left| \frac{v_c^2}{H} \sim \nu \frac{v_c}{S_T^2} \right|$

$$\frac{v_c}{H} \sim \frac{\nu}{S_T^2}$$

$$S_T^2 = \frac{H \nu}{v_c}$$

$$S_T = \frac{H \nu}{\alpha/H Ra_H^{1/2}} \Rightarrow H^2 Pr Ra_H^{-1/2}$$

$$S_T \sim H Pr^{1/2} Ra_H^{-1/4}$$

$$\frac{S_T}{S} = \frac{H Ra_H^{-1/4}}{H Pr^{1/2} Ra_H^{-1/4}}$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} =$$

$$u^* = \frac{u}{v_c} \quad ; \quad v^* = \frac{v}{v_c}$$

$$u = u^* v_c$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*}$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} =$$

$$g \beta_\infty (T_0 - T)$$

$$g \beta_\infty \frac{(T_0 - T)}{H}$$

When the

$$= 1 \rightarrow$$

$$\frac{S_T}{S} = \frac{H Ra_H^{-1/4}}{H Pr^{1/2} Ra_H^{-1/4}} \Rightarrow \underline{P_r^{-1/2}}$$

$$\underline{P_r^{-1/2}}$$

$$\boxed{S_T < S}$$

Forced vs Natural Convection

$$u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial x^2} + g \beta_\infty (T - T_\infty)$$

13/3/24

vertical (y-coord)
momentum balance
eqn.

$$u^* = \frac{u}{v_c} \quad ; \quad v^* = \frac{v}{v_c} \quad ; \quad x^* = \frac{x}{H}$$

$$\theta = \frac{T - T_\infty}{T_o - T_\infty}$$

$$u = u^* v_c \quad ; \quad v = v^* v_c \quad ; \quad y^* = \frac{y}{H}$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*} + \frac{v_c^2}{H} v^* \frac{\partial u^*}{\partial y^*} = \frac{v_c}{H^2} \nu \frac{\partial^2 u^*}{\partial x^{*2}} + g \beta_\infty (T_o - T_\infty) \theta$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = \frac{H}{v_c^2} \times \frac{v_c}{H^2} \nu \frac{\partial^2 u^*}{\partial x^{*2}} + \frac{H}{v_c^2} g \beta_\infty (T_o - T_\infty) \theta$$

If this
co-efficient is significantly
high then the buoyancy
force is significantly high.

for a given
system this
can't be controlled.

$$g \beta_\infty \frac{(T_o - T_\infty) H^3}{\nu^2} \times \frac{H^2 \nu^2}{v_c^2 H^3} \quad \left(\begin{array}{l} \text{Divide by } \nu^2 \text{ multiply by } \nu^2 \\ \text{To make Grashoff number.} \end{array} \right)$$

$$g \beta_\infty \frac{(T_o - T_\infty) H^3}{\nu^2} \times \frac{\nu^2}{v_c^2 H^2}$$

$$\sim \frac{Gr_o}{Re^2} \sim Ri = \text{Richardson Number}$$

When this Richardson Number is $\gg 1$, $\rightarrow$ lead to free convection
 $Re < 1 \rightarrow$ buoyancy effect neglected forced convection.
convection.

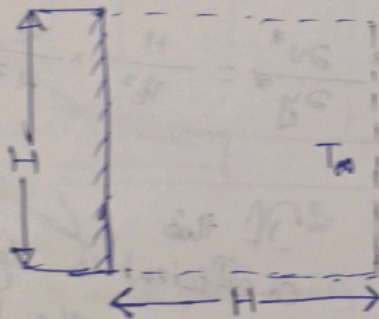
Integral Solution

$$\frac{d}{dy} \int_0^x u^2 dx = -v \left(\frac{\partial u}{\partial x} \right)_{x=0} + g\beta \int_0^x (T - T_\infty) dx \quad (1)$$

$$\frac{d}{dy} \int_0^x v (T_\infty - T) dx$$

Assume a velocity profile & Temperature profile.
 High Pr fluids & low Prandtl fluids (Note)
Similarity Solutions (From book)

Problems 1



$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = v \frac{\partial^2 v}{\partial x^2} + g\beta (\Delta T)$$

Scale of x for this system

$$x \sim H; \quad y \sim H; \quad v \sim v_c; \quad u \sim u_c.$$

$$u_c \frac{\partial v_c}{\partial H} + v_c \frac{\partial v_c}{\partial H} =$$

$$\frac{v_c^2}{H}$$

$$\frac{v_c^2}{H^2}$$

$$g\beta(\Delta T)$$

$$\frac{v_c^2}{H^2} \sim g\beta \Delta T$$

$$\Rightarrow v_c \sim \frac{g\beta \Delta T H^2}{\nu}$$

Chapter 5

Integral Solutions

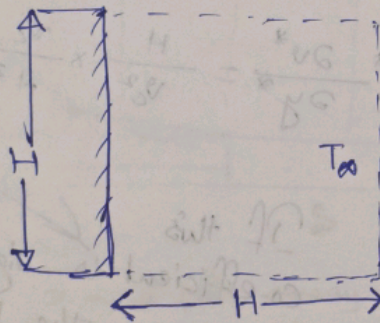
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$$\frac{d}{dy} \int_0^x v (T_\infty - T) dx$$

Assume a velocity profile & Temperature profile.
High Pr fluids & low Prandtl fluids (Note)

Similarity Solutions (From Book)

Problems 1



eqn

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \nu \frac{\partial^2 v}{\partial x^2} + g\beta (\Delta T)$$

scale of x for this system

$$x \sim H; \quad y \sim H; \quad v \sim v_c; \quad u \sim u_c.$$

$$u_c \frac{\partial v_c}{\partial H} + v_c \frac{\partial v_c}{\partial H} =$$

$$\frac{v_c^2}{H}$$

$$\frac{v_c^2}{H^2}$$

$$g\beta (\Delta T)$$

$$\frac{v_c^2}{H^2} \sim g\beta \Delta T$$

$$\Rightarrow v_c \sim \frac{g\beta \Delta T H^2}{\nu}$$

Chapter 5

equating inertia & friction to get Grashoff no.

$$\frac{\text{inertia}}{\text{friction}} \sim \frac{v_c^2/H}{\frac{\mu v_c}{H^2}} \sim \frac{v_c H}{\mu}$$

If we multiply this,

If we substitute v_c with its value we get Grashoff no.

Part 2

conduction ~ convection. (from energy equation)

$$\sim \propto \frac{\Delta T}{H^2} \sim \frac{v_c \Delta T}{H} \Rightarrow v_c \sim \frac{\alpha}{H}$$

2 eqn and only unknown is v_c , so we are getting two v_c .

Chapter 7

TBL

7.1, 7.2, 7.22 eqn, 7.4, 7.3, 7.41

7.69

Colburn analogy,

7.81, 7.82

Solve

upto

7.8

(Initial 15 problem) Reynolds Analogy

Chapter 4

ENC

4.1
4.3

Boussinesq approximation,

Scale Analysis, (4.3.1) High Pr
Table 4.1 (4.3.1) fluids

4.4 (Integral solution + similarity soln)

4.6 (Uniform Heat flux)

upto page 192

4.9 as a problem & solved.

4.10 described in different ways. (4.112)

4.11.2, 4.11.3,

Horizontal cylinder (Page 209)

5.1, 5.8

What is the regime (5.12)

5.2, Integral soln. X

2.49 (upto)

Shallow

Chapter 5

INC

$$\delta_T \sim \left(\frac{\alpha L}{u_{\infty}} \right)^{1/2} = \left(\frac{\alpha \nu L^2}{u_{\infty} \nu L} \right)^{1/2} = L \left(\frac{\alpha}{\nu} \right)^{1/2} \left(\frac{\nu}{u_{\infty} L} \right)^{1/2}$$

$$\frac{\delta_T}{L} \sim Pr^{-1/2} Re^{-1/2}$$

$$\delta_T \ll \delta$$

AHT (MC) [Natural Convection] 12/3/24

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial x^2} + g(\rho_a - \rho) \quad \text{--- ①}$$

$$\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = \alpha \frac{\partial^2 T}{\partial x^2}$$

Assumptions
 $\rho = \rho_a + \left. \frac{\partial \rho}{\partial T} \right|_{T=T_a} (T - T_a) + \text{higher order terms.}$

$$(\rho - \rho_a) = \left. \frac{\partial \rho}{\partial T} \right|_{T=T_a} (T - T_a) + \dots$$

Notes
 $(\rho - \rho_a) = -\beta_a \rho_a (T - T_a)$

$$(\rho_a - \rho) = \beta_a \rho_a (T - T_a)$$

Boussinesq Approximation,

Density is constant everywhere, except the location of driving force.

In the eqn ①,

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial x^2} + g(\rho_a - \rho)$$

← expresses as ρ_a

→ This is constant

$$\rho \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = \alpha \frac{\partial^2 T}{\partial x^2}$$

Volume Expansion Co-efficient

$$\beta = \frac{1}{\nu} \left. \frac{\partial \nu}{\partial T} \right|_T$$

$$= \frac{1}{(\nu_p)} \left. \frac{\partial (\nu_p)}{\partial T} \right|_T$$

$$= \frac{1}{\rho} \left. \frac{\partial \rho}{\partial T} \right|_T$$

$$\beta_a = -\frac{1}{\rho_a} \left. \frac{\partial \rho}{\partial T} \right|_{T=T_a}$$

$$-\beta_a \rho_a = \left. \frac{\partial \rho}{\partial T} \right|_{T=T_a}$$

→ This will be the ρ_a of Temp. q/c to Boussinesq Approx.

Ans

$$\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}\right) = \nu \frac{\partial^2 v}{\partial x^2} + [g \beta_0 (T - T_a)]$$

$$\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}\right) = \alpha \frac{\partial^2 T}{\partial x^2}$$

Using Scaling Analysis we can solve this

$$\begin{aligned} Q &= A_{max} \times h_{max} \\ h &\sim \frac{k}{S_T} \text{ (scaling)} \end{aligned}$$

2 ~~at~~ $y \sim H, x \sim S_T, u \sim u.$
 $v \sim v_c$ ($v_c \approx$ characteristic velocity)

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\sim \frac{u}{S_T} \sim \frac{v_c}{H} \Rightarrow \boxed{u \sim \frac{v_c S_T}{H}} \quad \text{1}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\begin{aligned} \sim u \frac{\Delta T}{S_T} &\sim v_c \frac{\Delta T}{H} & \sim \alpha \frac{\Delta T}{S_T^2} \\ & & \sim \alpha \frac{\Delta T}{S_T^2} \end{aligned} \Rightarrow \boxed{v_c = \frac{\alpha H}{S_T^2}} \quad \text{2}$$

$$\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}\right) = \nu \frac{\partial^2 v}{\partial x^2} + g \beta (T - T_a)$$

$$\begin{aligned} \sim u \frac{v_c}{S_T} &\sim v_c \frac{v_c}{H} & \sim \nu \frac{v_c}{S_T^2} &\sim g \beta (\Delta T) \\ \sim \frac{v_c^2}{H} & & \sim \nu \frac{v_c}{S_T^2} & \sim g \beta (\Delta T) \end{aligned}$$

Inertial force.

Frictional/viscous

Buoyancy

The Assumption for

natural convection is

buoyancy force has to be there

$$\frac{\text{Friction}}{\text{Buoyancy}} = \frac{\nu \frac{v_c}{S_T^2}}{g \beta \Delta T}$$

Substituting

$$\sim \frac{1}{RaH} \left(\frac{H}{\Delta T}\right)^4$$

Frictional force

$$\frac{\text{Inertia}}{\text{Buoyancy}} = \frac{\frac{v_c}{H}}{g \beta \Delta T}$$

For high Prandtl

$$RaH^{-1} \left(\frac{H}{S_T}\right)^4$$

$$\boxed{S_T = H}$$

Scale of $H, h \sim$

$$\boxed{Nu \sim R}$$

$$\frac{\text{Friction}}{\text{Buoyancy}} = \frac{\frac{\nu_c}{S_T^2}}{g \beta \Delta T} = \frac{\frac{\alpha H}{S_T^4}}{g \beta \Delta T}$$

$$= \left[\frac{\alpha}{g \beta \Delta T H^3} \right] \times \left(\frac{H^4}{S_T^4} \right)$$

$$= \frac{1}{Ra_H} \times \left(\frac{H}{S_T} \right)^4 \rightarrow \text{Rayleigh number}$$

Substituting

$$\sim \frac{1}{Ra_H} \left(\frac{H}{\Delta T} \right)^4$$

Frictional force

$$\sim \frac{1}{Ra_H} \times \frac{1}{Pr} \times \left[\frac{H}{S_T} \right]^4 \sim 1.$$

Inertial force

$$\frac{\text{Inertia}}{\text{Buoyancy}} = \frac{\frac{\nu_c^2}{H}}{g \beta \Delta T} = \frac{\frac{\alpha^2 H^2}{S_T^4} \times \frac{1}{H}}{g \beta \Delta T}$$

$$= \frac{\alpha}{g \beta \Delta T H^3} \times \left[\frac{\alpha}{\nu} \right] \times \left[\frac{H}{S_T} \right]^4$$

$$= \frac{1}{Ra_H} \times \frac{1}{Pr} \times \left[\frac{H}{S_T} \right]^4$$

Hi for high Prandtl number fluid

$$Ra_H^{-1} \left(\frac{H}{S_T} \right)^4 \sim 1.$$

$$S_T = H Ra_H^{-1/4}$$

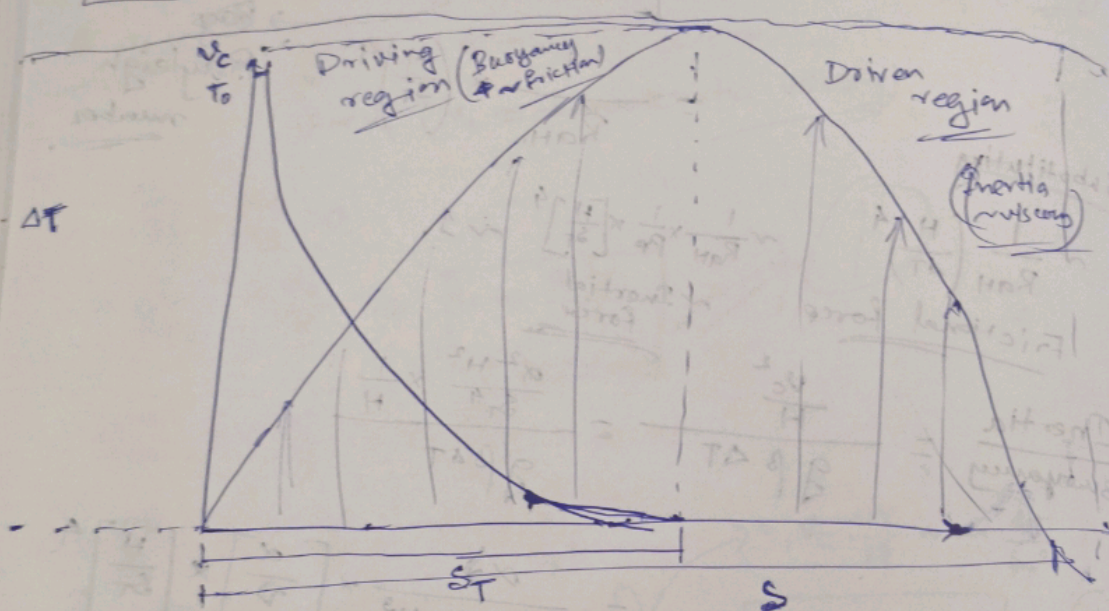
$$\text{Scale of } H, h \sim \frac{\kappa}{S_T} \sim \frac{\kappa}{H} Ra_H^{1/4}$$

$$Nu \sim Ra_H^{1/4}$$

Energy Equation

$$v_c \sim \frac{\alpha H}{S_T^2}$$

$$v_c \sim \frac{\alpha}{H} Ra_H^{1/2}$$



Inertia $\rightarrow$ Driving the fluid $x \rightarrow$ (far away from wall)
 viscous $\rightarrow$ Resist the flow of fluid.

Driven region $\left| \frac{v_c^2}{H} \sim \nu \frac{v_c}{S_T^2} \right|$

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$$S_T^2 = \frac{H\nu}{v_c}$$

$$S_T = \frac{H\nu}{\alpha/H Ra_H^{1/2}} \Rightarrow H^2 Pr Ra_H^{-1/2}$$

$$S_T \sim H Pr^{1/2} Ra_H^{-1/4}$$

$$\frac{S_T}{S} = \frac{H Ra_H^{-1/4}}{H Pr^{1/2} Ra_H^{-1/4}}$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} =$$

$$u^* = \frac{u}{v_c} \quad ; \quad v =$$

$$u = u^* v_c$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*}$$

$$u^* \frac{\partial v^*}{\partial x^*} + v$$

$$g \beta_\Delta (T_o$$

$$g \beta_\Delta \frac{(T_o$$

When the

$$= 1 \rightarrow$$